

HAFIZ MUHAMMAD HUSSAIN SOHAIL

Mechanical Engineering Student | HVAC, Energy Systems & Design/Manufacturing

Lahore, Pakistan | +92 317 4370001 | hafizhussain3663@gmail.com | linkedin.com/in/hafiz-muhammad-hussain-sohail-6a98b0403

PROFESSIONAL SUMMARY

Mechanical Engineering undergraduate (CGPA 3.67/4.0) at UET Lahore with hands-on project experience in thermal systems, fluid mechanics, HVAC design, and manufacturing processes. Completed nine industry-recognized certifications in 2026 covering aerospace, robotics, and business analysis, including Forage simulations with British Airways, GE Aerospace, and Johnson & Johnson, plus HP LIFE and Deloitte WorldClass. Co-Director of a 50+ member ASHRAE student chapter. Proficient in AutoCAD, SolidWorks, MATLAB, and Python. Seeking an internship or graduate trainee role in HVAC, energy, or design engineering.

EDUCATION

B.Sc. Mechanical Engineering: CGPA 3.67 / 4.0

Sep 2024 – Jul 2028

University of Engineering & Technology (UET), Lahore

- Relevant Coursework: Fluid Mechanics, Thermodynamics, Manufacturing Processes, Heat Transfer, Machine Design, Engineering Materials, Engineering Project Management
- Recipient, Prime Minister Youth Laptop Award, awarded for outstanding academic performance

FSc Pre-Engineering

Aug 2022 – May 2024

KIPS College, Rahim Yar Khan: High Achiever Certificate

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- SolidWorks Certification, ASHRAE Student Chapter
- Engineering Job Simulation, British Airways (via Forage)
- Robotics and Controls Job Simulation, Johnson & Johnson MedTech (via Forage)
- Explore Engineering Job Simulation, GE Aerospace (via Forage)
- Effective Leadership, Deloitte WorldClass
- Profit and Loss, HP LIFE / HP Foundation
- Critical Thinking in the AI Era, HP LIFE
- Business Communications, HP LIFE
- PHYS102: Introduction to Electromagnetism, Saylor Academy
- FAA Safety Team ALC, FAA Aviation Safety: Improving Flight Safety with uAvionix AV-30/AV-HSI/AV-APA, and Management of Risk from a NASA Associate Director

LEADERSHIP & PROFESSIONAL EXPERIENCE

Co-Director, Technical Team (ASHRAE Student Chapter, UET Lahore)

Jun 2026 – Present

- Co-lead a 50+ member technical team; coordinate with faculty advisors to align chapter activities with ASHRAE's global engineering competency standards
- Direct technical workshops on ASHRAE standards, building energy systems, and sustainable HVAC design principles

Event Organizer, Integration Bee Competition (ASHRAE Student Chapter, UET Lahore)

May 2026

- Designed and executed a campus-wide mathematics competition modeled on MIT's Integration Bee, attracting 40+ student participants
- Managed end-to-end event logistics: problem curation, bracket design, scoring system, and day-of coordination

Donation Co-Head (RYK Kaar-e-Kamal)

Mar 2026 – May 2026

- Coordinated fundraising campaigns and donor outreach, securing resources for community development programs
- Organized a Ramadan Food Distribution Drive, managing volunteer allocation and budget tracking for essential food supplies to 100+ families

Graphic Designer (IMechE, ASHRAE & Freedom Fighters KSK, UET Lahore)

Jan 2025 – Present

- Produce technical and promotional design assets for three concurrent engineering organizations, maintaining consistent branding across digital platforms
- Design event posters, banners, and technical reports supporting ASHRAE chapter activities and IMechE competitions

Buffer Leader (UET Lahore Kaar-e-Kamal)

Oct 2025 – Jan 2026

- Led two teams of incoming members through onboarding into the organization

ENGINEERING PROJECTS

Robotic Arm with Mechanical Claw, SolidWorks Design Project, UET Lahore

Apr 2026 – Jun 2026

- Designed a multi-jointed robotic arm with a mechanical claw end-effector in SolidWorks, modeling linkages, joints, and gripper mechanism
- Built assembly and motion studies to validate range of motion, clearances, and claw actuation before physical build

Desiccant Evaporative Cooler, Class Project, UET Lahore

Apr 2026 – Jun 2026

- Handled all electrical work for a desiccant evaporative cooler system, including wiring and component selection for the cooling circuit and heating element
- Assembled the complete circuit, integrating fan motor, controls, and wiring into a fully functional prototype

Thermal Power Plant Analysis (Coal-Fired Systems), Academic Research, UET Lahore

Apr 2026 – May 2026

- Conducting thermodynamic analysis of coal-fired power cycles across four coal grades: anthracite, bituminous, sub-bituminous, and lignite
- Applying Dulong's formulation and energy balance equations to benchmark mass required, emissions, and fuel consumption against calorific value

Zero-Head Turbine Design & Energy Analysis, Fluid Mechanics Lab, UET Lahore

Apr 2026 – May 2026

- Engineering a low-head hydraulic turbine to extract usable energy from near-zero hydraulic head conditions for rural electrification
- Built CAD model and physical prototype in AutoCAD; produced full OEL performance report

Bernoulli's Equation Analysis, Fluid Mechanics Research

Oct 2025 – Nov 2025

- Developed an analytical model incorporating Darcy-Weisbach friction losses and minor loss coefficients into Bernoulli's equation for pipe flow
- Validated theoretical predictions against experimental lab data, achieving under 5% deviation in pressure-drop calculations

Parametric Machining Analysis, Grade-V Titanium

Dec 2025 – Jan 2026

- Analyzed cutting speed, feed rate, and depth of cut effects on tool wear, cutting forces, and surface finish for Ti-6Al-4V
- Benchmarked results against ISO surface roughness standards; identified parameter combinations reducing tool wear by an estimated 20%

Casting Process Analysis, Spur Gear, Manufacturing Engineering Project

Dec 2025 – Jan 2026

- Evaluated material trade-offs (cast iron, steel, aluminum alloys) for spur gear casting across mechanical properties, machinability, and cost
- Modeled shrinkage and porosity defect risk using solidification principles; built dimensioned CAD model in AutoCAD

Self-Opening Window Using Refrigerant Actuation, Thermodynamics Design Project

Oct 2024 – Nov 2024

- Designed a passive HVAC-adjacent ventilation mechanism using refrigerant thermal expansion for automatic actuation without electrical input
- Compared R-134a, R-22, and CO₂ as working fluids on actuator force, thermal response time, and environmental impact

TECHNICAL SKILLS

- Engineering Software: AutoCAD, SolidWorks, MATLAB, Origin Pro
- Programming: Python, HTML, CSS
- Core Engineering: Thermodynamic Analysis, Fluid Mechanics, Heat Transfer, HVAC Systems, Manufacturing Processes
- Productivity: Advanced Excel, Word, PowerPoint, Technical Report Writing
- Languages: English (Professional Proficiency), Urdu (Native)